Problem Set 6 - Lists

Hello student,

the goal of this problem-set is to familiarize with **lists** and how to manipulate them.

Assignment 6.1 - Vector Multiplication (optional)

Write a program that asks the user for a vector (x,y,z components), multiplies it by a number, also inserted by the user, and store it into a new variable. The program must then print the original vector, multiplication value and resulting vector. Use lists to solve the problem.

The output **must** look like this:

```
Initial vector:
    x,y,z => 2,1,0

Value for multiplication: 2

Resulting vector:
    x,y,z => 4,2,0
```

Assignment 6.2 - Sum and mean (optional)

Write a program that asks the user for a repetitive input of integer numbers. The request should stop when the user inserts the number **0**.

Then the program should calculate:

- The mean of all the numbers
- The sum of all the numbers
- The quantity of positive and negative numbers
- The percentage of positive and negative numbers

The output **must** be like this (in red the input by the user):

```
1. number: 1
2. number: -4
3. number: 2
4. number: -3
5. number: 0

Inserted 0, finishing number requests.

- The mean of all the numbers is -1.0
- The sum of all the numbers is -4
- The inserted negative numbers are: -4,-3
- The amount of negative numbers is 2 of 4 => [50%]
- The inserted positive numbers are: 1,2
- The amount of positive numbers is 2 of 4 => [50%]
```

Assignment 6.3 - Remove duplicate (optional)

Write a program that asks the user for 6 numbers between 6 and 666 included.
The number must be stored in a list only if is not a duplicate of an already saved number and only if in the required range. At the end print the 6 stored number in ascending order.

The output **must** look like this (in red the input by the user):

```
Insert a number: 312
Insert a number: 199
Insert a number: 312
Invalid number [312]. It was already inserted! Insert another number: 42
Insert a number: 4
Invalid number [4]. It is out of range! Insert another number: 333
Insert a number: 699
Invalid number [699]. It is out of range! Insert another number: 526
Insert a number: 601

Numbers => 42, 199, 312, 333, 526, 601
```

Assignment 6.4 - Find the word (optional)

Write a program that asks the user for 10 words that should be stored in a list. At the end the program should look, inside the list, for a word chosen by the user.

The output **must** look like this (in red the input by the user):

```
Word 1 = ant
Word 2 = tiger
Word 3 = Elephant
Word 4 = Rat
Word 5 = horse
Word 6 = cat
Word 7 = Dog
Word 8 = Lion
Word 9 = maMMuT
Word 10 = Panda

Insert the word to find: maMMuT

The word found is the number 9 and is at the index 8 of the list.
```

Assignment 6.5 - List manipulation (optional)

Write a program that asks the user for 10 words and store them in a list.

The program must then print the list in alphabetical order both ascending and descending, in reverse order and as originally inserted.

At the end, the program must print for each unique word in the list how many time it was inserted.

For example (user input in red):

```
Word: Ant
Word: Lion
Word: Elephant
Word: Rat
Word: Horse
Word: Rat
Word: Dog
Word: Lion
Word: Panda
Word: Panda
```

The output **must** then look like:

```
Alphabetical order:
'Ant'-'Dog'-'Elephant'-'Horse'-'Lion'-'Lion'-'Panda'-'Panda'-'Rat'-'Rat'

Reversed alphabetical order:
'Rat'-'Rat'-'Panda'-'Panda'-'Lion'-'Lion'-'Horse'-'Elephant'-'Dog'-'Ant'

Reversed order:
'Panda'-'Panda'-'Lion'-'Dog'-'Rat'-'Horse'-'Rat'-'Elephant'-'Lion'-'Ant'

Original order:
'Ant'-'Lion'-'Elephant'-'Rat'-'Horse'-'Rat'-'Dog'-'Lion'-'Panda'-'Panda'

---- Count ----

Rat: 2
Panda: 2
Lion: 2
Horse: 1
Elephant: 1
Dog: 1
Ant: 1
```